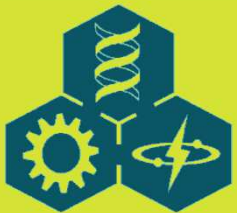
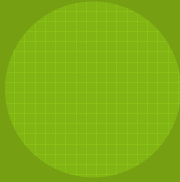


LECTURE 10:

STOCHASTIC MODELS





OUTLINE

10.1 There's Nothing Like a Random World

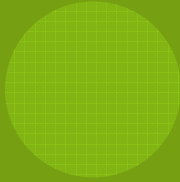
10.2 Random Numbers

10.3 Sampling Strategies

10.4 Applications to Differential Equations

10.5 Markov Processes





10.1 THERE'S NOTHING LIKE A RANDOM WORLD

- ⊙ Biological systems (like many other systems) are subject to apparently random fluctuations. That is, either the state variables themselves or the parameters are perturbed at random times and by random amounts.
- ⊙ Repeatedly simulating random models allows us to estimate characteristics of the probabilistic model response (e.g., the distribution's dispersion and central tendency). This process is called *Monte Carlo simulation*.



10.1 THERE'S NOTHING LIKE A RANDOM WORLD

- © There are three broad areas in which probabilistic models and Monte Carlo simulation are useful in biological simulation:



Statistical Hypotheses



Differential and Difference Equations



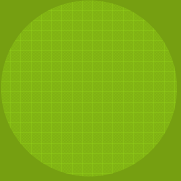
Markov Processes



10.1 THERE'S NOTHING LIKE A RANDOM WORLD

- ◎ When faced with random variation in systems, modelers have two fundamental choices to make. They can either ignore these random changes and model mean behavior, or they can incorporate randomness by constructing stochastic models and couching predictions in terms of probable outcomes.
- ◎ In this chapter, we will discuss the following topics:
 - (1) the mechanics of generating and using computer-generated random numbers in simulation,
 - (2) simulating stochastic differential equations, and
 - (3) Markov chains.





10.2 RANDOM NUMBERS

- ◎ The reason for unpredictability is that a large number of unknown physical events are interacting in complex ways.
- ◎ The combinations of events and interactions are so large that from the human perspective of limited knowledge, the outcome is unpredictable.
- ◎ At times, we want to select numbers (also called *deviates*) from normal, exponential, gamma, or other distributions. Or, we may wish to choose numbers from empirical distributions.
- ◎ For a large class of distributions, if we can generate random numbers from a uniform distribution, then it is a rather simple matter to use these numbers to obtain a deviate from the desired distribution.



10.2 RANDOM NUMBERS

Generating Uniform Random Numbers

- For the methods that use recursive equations, the last number produced is used to calculate the next. To emphasize the deterministic origins of the numbers, we call them *pseudo-random numbers*.
- One such operation that lies at the heart of many algorithms is the mod or modulus arithmetic operation. $(y \bmod x)$ produces the integer remainder obtained by dividing y by x .
- For example,

$$x_{n+1} = x_n^2 \bmod 31417$$

If we begin with $x_0 = 123$, we generate the following sequence of numbers: 123, 15129, 13796, 5430, 15754, 25633, 26968, 891, 8456, 30261, 16822.



10.2 RANDOM NUMBERS

Generating Uniform Random Numbers

- ⦿ There are several critical characteristics of good algorithms.
 1. They should produce long sequences before repeating.
 2. They should be fast.
 3. They should reproduce the major components of the desired distribution (mean, variance, skew, distribution at the tails, etc.).
 4. They should minimize the use of computer memory.



10.2 RANDOM NUMBERS

Generating Uniform Random Numbers

- Almost all modern compilers provide a built-in function that returns a random number from a uniform distribution. Although it varies among compiler manufacturers, the *linear congruential* method is most commonly used. It is the recursive function:

$$U_{i+1} = (aU_i + c) \bmod m \quad (10.1)$$

where a , c , and m are machine-dependent constants chosen to produce a good fit to a uniform distribution. For example, on an IBM mainframe computer, $a = 3\,141,592,69$; $c = 453,806,245$; and $m = 2^{31}$. If $c = 0$, it is a multiplicative congruential method.



10.2 RANDOM NUMBERS

- ◎ Generating Uniform Random Numbers
 - ◎ Modern implementations now frequently use $m = 2^{32} - 1$. For most compilers in which the longest integer is 32 bits, the period is close to $2^{32} \approx 4 \times 10^9$.
 - ◎ The shuffling method (Press et al. 1992) gives a period of about 2×10^{18} . In this method, a computed random number r_n is not output as the n -th random number, but rather at a randomized later call.
 - ◎ Since the pseudo-random method is a recursive algorithm, the initial random *seed* (U_0 in Eq. 10.1) must be supplied before a sequence can be produced. All random generator libraries supply a function for initializing the sequence.

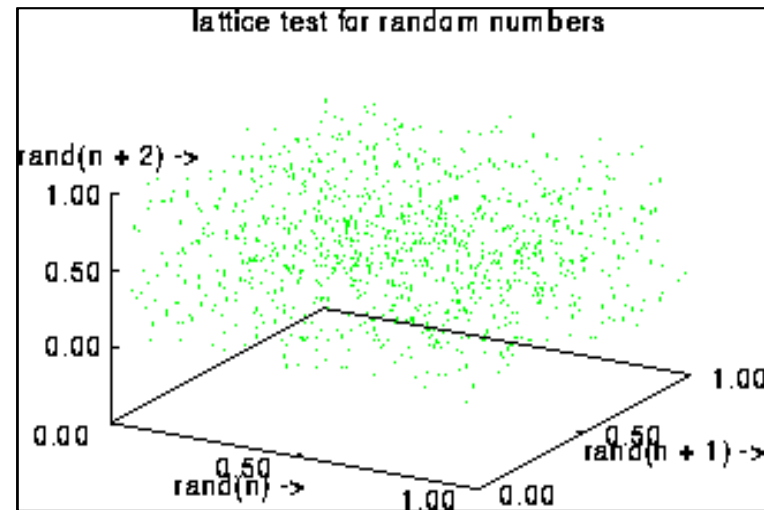
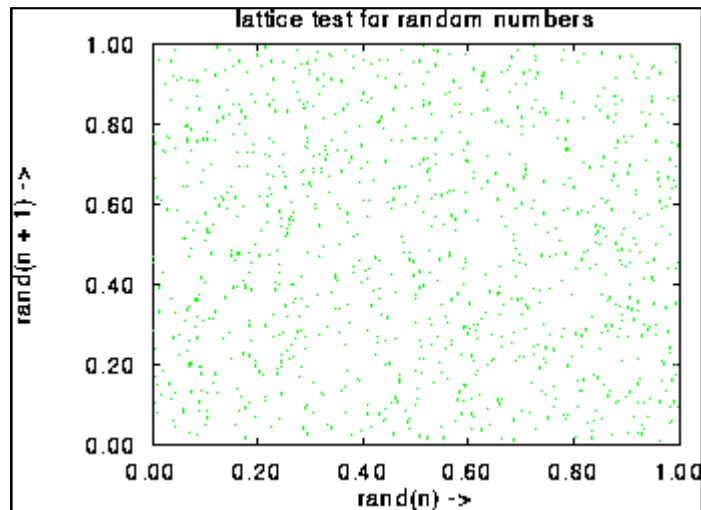


10.2 RANDOM NUMBERS

Generating Uniform Random Numbers

Test for random number generators:

1. Check for frequency distribution.
2. Chi-square test.
3. Lattice test (scatter diagram).



10.2 RANDOM NUMBERS

Generating Normal Deviates

- The Box-Muller method involves combining two uniform random numbers (U_1, U_2 , obtained from 2 separate calls to the uniform generator) to produce two random numbers from a normal distribution (z_1, z_2) having a mean of 0 and a standard deviation of 1.0 [i.e., $N(0, 1)$]:

$$z_1 = \sqrt{-2\ln(U_1)} \cos(2\pi U_2) \quad z_2 = \sqrt{-2\ln(U_1)} \sin(2\pi U_2)$$

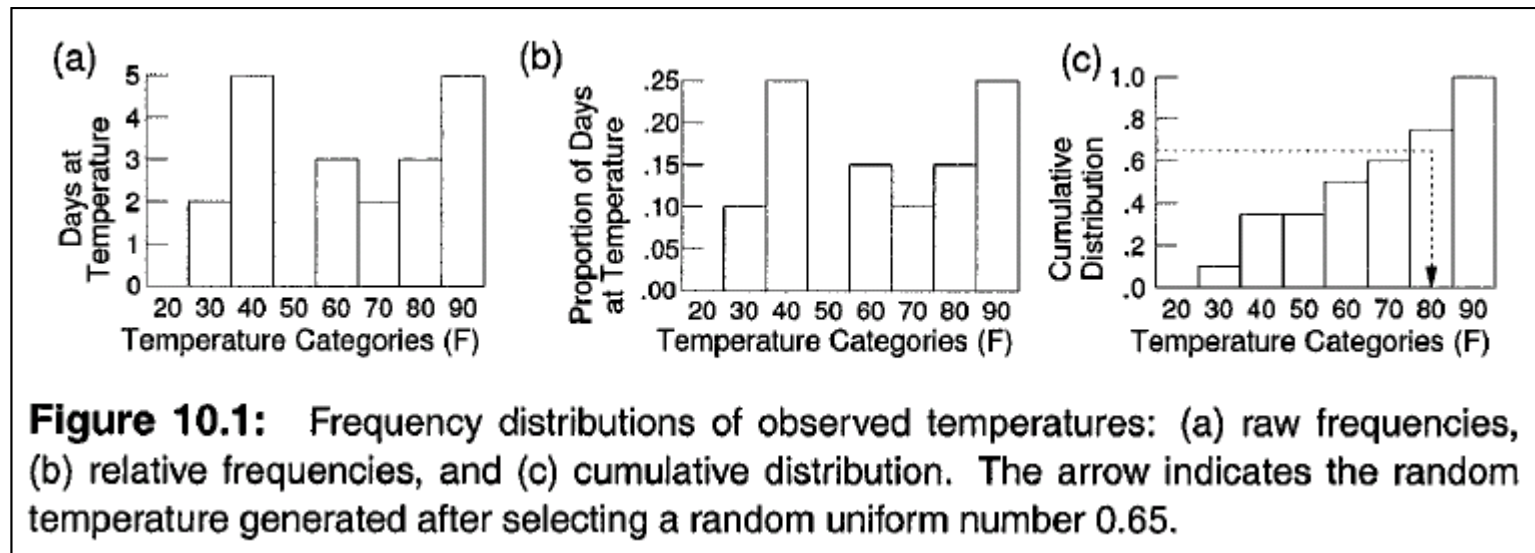
- To convert a standardized random deviate (z_i above) to a deviate from another normal distribution with standard deviation s and mean m , use $y = sz + m$.



10.2 RANDOM NUMBERS

◎ Inverse Cumulative Methods

- ◎ To sample from the distribution, we use table look-up. The random deviate is the mid-point of the bin. Notice that the initial width of the categories determines the resolution of the deviate generated.



10.2 RANDOM NUMBERS

◎ Inverse Cumulative Methods

- ◎ An example of the wrapped Cauchy distribution

The probability density function

$$f(\phi) = \frac{1 - \rho^2}{2\pi[1 + \rho^2 - 2\rho \cos(\phi - \theta)]} \quad (10.2)$$

The cumulative distribution function is the integral of the pdf

$$\begin{aligned} F(\omega) &= \frac{1 - \rho^2}{2\pi} \int \frac{dx}{1 + \rho^2 - 2\rho \cos(\omega)} \\ &= \frac{1}{\pi} \arctan\left(\frac{1 - \rho}{1 + \rho} \tan\left[\frac{\omega}{2}\right]\right) + C \end{aligned} \quad (10.3)$$

$$F^{-1}(\omega) = \phi = \theta + 2 \arctan\left(\frac{1 - \rho}{1 + \rho} \tan(\pi(U(0,1) - 0.5))\right) \quad (10.4)$$



10.2 RANDOM NUMBERS

◎ Inverse Cumulative Methods

◎ An example of the wrapped Cauchy distribution

To summarize, the method to sample a deviate x is:

1. Determine the pdf of x [$f(x)$].
2. Integrate to get the cdf [$F(x)$].
3. Determine the constant of integration at $F(x) = 1$ and/or $F(x) = 0$.
4. Set $F(x)$ to be a value from the uniform distribution [$U(0, 1)$].
5. Invert $F(x)$ and solve for x .



10.2 RANDOM NUMBERS

- ◎ Methods for Other Distributions
 - ◎ Methods for the Cauchy, log-normal, exponential, gamma, F, and Weibull continuous distributions, and the binomial, Poisson, hypergeometric, and negative binomial discrete distributions can be found in numerical software packages [e.g., the GNU Scientific Library (GSL), International Mathematical and Statistical Library (IMSL), Numerical Algorithm Group (NAG), Mathematica, etc.] or in more advanced texts.



10.2 RANDOM NUMBERS

◎ Methods for Other Distributions

◎ MATLAB functions

Syntax

`Y = random(name,A)`

`Y = random(name,A,B)`

`Y = random(name,A,B,C)`

`Y = random(...,m,n,...)`

`Y = random(...,[m,n,...])`

Examples

1. Generate a 2-by-4 array of random values from the normal distribution with mean 0 and standard deviation 1:

`x1 = random('Normal',0,1,2,4)`

2. Generate a single random value from Poisson distributions with rate parameters 1, 2, ..., 6, respectively:

`x2 = random('Poisson',1:6,1,6)`



10.2 RANDOM NUMBERS

Methods for Other Distributions

MATLAB functions

name	Distribution	Input Parameter A	Input Parameter B	Input Parameter C
'beta' or 'Beta'	Beta Distribution	a	b	—
'bino' or 'Binomial'	Binomial Distribution	n: number of trials	p: probability of success for each trial	—
'chi2' or 'Chisquare'	Chi-Square Distribution	v: degrees of freedom	—	—
'exp' or 'Exponential'	Exponential Distribution	μ : mean	—	—
'ev' or 'Extreme Value'	Extreme Value Distribution	μ : location parameter	σ : scale parameter	—
'f' or 'F'	F Distribution	v1: numerator degrees of freedom	v2: denominator degrees of freedom	—
'gam' or 'Gamma'	Gamma Distribution	a: shape parameter	b: scale parameter	—
'gev' or 'Generalized Extreme Value'	Generalized Extreme Value Distribution	k: shape parameter	σ : scale parameter	μ : location parameter
'gp' or 'Generalized Pareto'	Generalized Pareto Distribution	k: tail index (shape) parameter	σ : scale parameter	μ : threshold (location) parameter
'geo' or 'Geometric'	Geometric Distribution	p: probability parameter	—	—
'hyge' or 'Hypergeometric'	Hypergeometric Distribution	M: size of the population	K: number of items with the desired characteristic in the population	n: number of samples drawn
'logn' or 'Lognormal'	Lognormal Distribution	μ	σ	—
'nbin' or 'Negative Binomial'	Negative Binomial Distribution	r: number of successes	p: probability of success in a single trial	—



10.2 RANDOM NUMBERS

- Methods for Other Distributions
 - MATLAB functions

name	Distribution	Input Parameter A	Input Parameter B	Input Parameter C
'ncf' or 'Noncentral F'	Noncentral F Distribution	$v1$: numerator degrees of freedom	$v2$: denominator degrees of freedom	δ : noncentrality parameter
'nct' or 'Noncentral t'	Noncentral t Distribution	v : degrees of freedom	δ : noncentrality parameter	—
'ncx2' or 'Noncentral Chi-square'	Noncentral Chi-Square Distribution	v : degrees of freedom	δ : noncentrality parameter	—
'norm' or 'Normal'	Normal Distribution	μ : mean	σ : standard deviation	—
'poiss' or 'Poisson'	Poisson Distribution	λ : mean	—	—
'rayl' or 'Rayleigh'	Rayleigh Distribution	b : scale parameter	—	—
't' or 'T'	Student's t Distribution	v : degrees of freedom	—	—
'unif' or 'Uniform'	Uniform Distribution (Continuous)	a : lower endpoint (minimum)	b : upper endpoint (maximum)	—
'unid' or 'Discrete Uniform'	Uniform Distribution (Discrete)	N : maximum observable value	—	—
'wbl' or 'Weibull'	Weibull Distribution	a : scale parameter	b : shape parameter	—



10.2 RANDOM NUMBERS

- ◎ Multivariate Distributions
 - ◎ If variables are correlated, then the distribution is multivariate and we can not draw the deviates independently. The method to use depends on the underlying distribution.
 - ◎ The sampling distribution of a function is portrayed by its variance-covariance matrix. This square matrix must be considered when drawing deviates from a multivariate distribution.



10.2 RANDOM NUMBERS

◎ Multivariate Distributions

- ◎ If the distribution of n variates is normal, then the following algorithm returns a deviate for each of the variables. (1) Select n deviates ($\mathbf{z}$) from the standard normal distribution using the Box-Mueller method (or equivalent). (2) Convert the n standard deviates into physical deviates with the relation $\mathbf{y} = \mathbf{m} + \mathbf{S}\mathbf{z}$, where $\mathbf{m}$ is the vector of variable means and $\mathbf{S}$ is a square matrix derived from the variance-covariance matrix ($\mathbf{V}$).
- ◎ The following relationship holds:

$$\mathbf{V} = \mathbf{S}\mathbf{S}'$$

When $n > 2$, we use software to generate the Cholesky decomposition to obtain $\mathbf{S}$.



10.2 RANDOM NUMBERS

◎ Multivariate Distributions

- ◎ When $n = 2$, we can easily do it by hand as follows.

$$\mathbf{V} = \mathbf{S}\mathbf{S}'$$

$$\begin{pmatrix} \sigma_1^2 & \sigma_{12} \\ \sigma_{21} & \sigma_2^2 \end{pmatrix} = \begin{pmatrix} s_{11} & 0 \\ s_{21} & s_{22} \end{pmatrix} \begin{pmatrix} s_{11} & s_{21} \\ 0 & s_{22} \end{pmatrix}$$

$$s_{11} = \sqrt{\sigma_1^2}$$

$$s_{21} = \sigma_{12} / \sqrt{\sigma_1^2}$$

$$s_{12} = 0$$

$$s_{22} = \sqrt{\sigma_2^2 - (\sigma_{12}^2 / \sigma_1^2)}$$

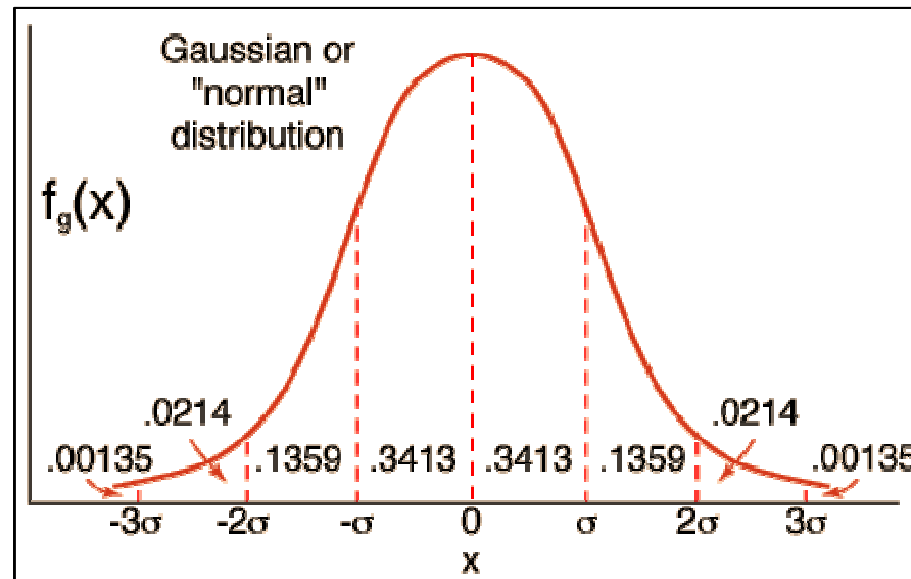


10.2 RANDOM NUMBERS

◎ Normal Distribution Function

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left[-\frac{(x - \mu)^2}{2\sigma^2}\right]$$

$$E(x) = \mu \quad \text{Var}(x) = \sigma^2$$



10.2 RANDOM NUMBERS

◎ Lognormal Density Function

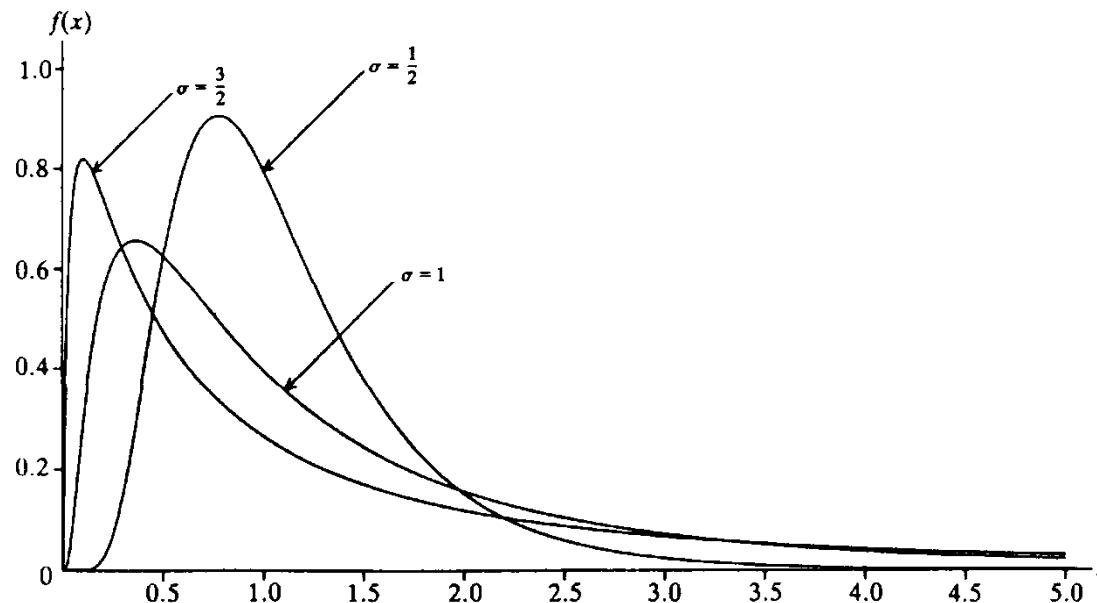
Lognormal	$LN(\mu, \sigma^2)$
Possible applications	Time to perform some task [density takes on shapes similar to $\text{gamma}(\alpha, \beta)$ and $\text{Weibull}(\alpha, \beta)$ densities for $\alpha > 1$, but can have a large “spike” close to $x = 0$ that is often useful]; quantities that are the product of a large number of other quantities (by virtue of central limit theorems)
Density (see Fig. 6.6)	$f(x) = \begin{cases} \frac{1}{x\sqrt{2\pi\sigma^2}} \exp \frac{-(\ln x - \mu)^2}{2\sigma^2} & \text{if } x > 0 \\ 0 & \text{otherwise} \end{cases}$
Distribution	No closed form
Parameters	Shape parameter $\sigma > 0$, scale parameter $\mu \in (-\infty, \infty)$
Range	$[0, \infty)$
Mean	$e^{\mu + \sigma^2/2}$
Variance	$e^{2\mu + \sigma^2}(e^{\sigma^2} - 1)$
Mode	$e^{\mu - \sigma^2}$
MLE	$\hat{\mu} = \frac{\sum_{i=1}^n \ln X_i}{n}, \quad \hat{\sigma} = \left[\frac{\sum_{i=1}^n (\ln X_i - \hat{\mu})^2}{n} \right]^{1/2}$
Comments	1. $X \sim LN(\mu, \sigma^2)$ if and only if $\ln X \sim N(\mu, \sigma^2)$. Thus, if one has data $X_1, X_2, \dots, X_n$ that are thought to be lognormal, the logarithms of the data points, $\ln X_1, \ln X_2, \dots, \ln X_n$, can be treated as normally distributed data for purposes of hypothesizing a distribution, parameter estimation, and goodness-of-fit testing



10.2 RANDOM NUMBERS

◎ Lognormal Density Function

2. As $\sigma \rightarrow 0$, the lognormal distribution becomes degenerate at e^μ .
Thus, lognormal densities for small σ have a sharp peak at the mode
3. $\lim_{x \rightarrow 0} f(x) = 0$, regardless of the parameter values



$LN(0, \sigma^2)$ density functions.



10.2 RANDOM NUMBERS

◎ Uniform Distribution Function

Uniform	$U(a, b)$
Possible applications	Used as a “first” model for a quantity that is felt to be randomly varying between a and b but about which little else is known. The $U(0,1)$ distribution is essential in generating random values from all other distributions (see Chaps. 7 and 8)
Density (see Fig. 6.1)	$f(x) = \begin{cases} \frac{1}{b-a} & \text{if } a \leq x \leq b \\ 0 & \text{otherwise} \end{cases}$
Distribution	$F(x) = \begin{cases} 0 & \text{if } x < a \\ \frac{x-a}{b-a} & \text{if } a \leq x \leq b \\ 1 & \text{if } b < x \end{cases}$
Parameters	a and b real numbers with $a < b$; a is a location parameter, $b - a$ is a scale parameter
Range	$[a, b]$
Mean	$\frac{a+b}{2}$
Variance	$\frac{(b-a)^2}{12}$



10.2 RANDOM NUMBERS

◎ Uniform Distribution Function

Mode

MLE

Comments

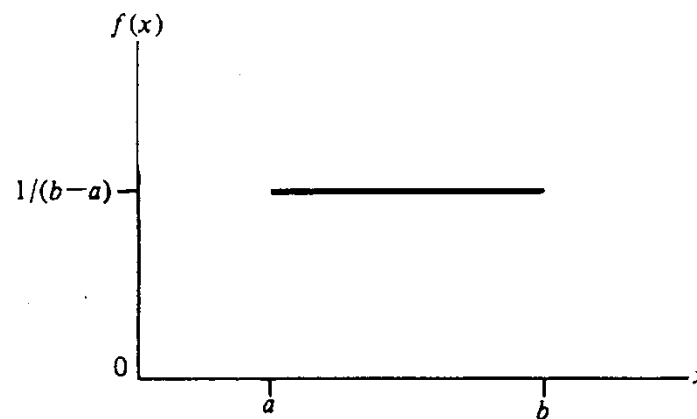
Does not uniquely exist

$$\hat{a} = \min_{1 \leq i \leq n} X_i, \hat{b} = \max_{1 \leq i \leq n} X_i$$

1. The $U(0,1)$ distribution is a special case of the beta distribution (when $\alpha_1 = \alpha_2 = 1$)
2. If $X \sim U(0,1)$ and $[x, x + \Delta x]$ is a subinterval of $[0,1]$ with $\Delta x \geq 0$,

$$P(X \in [x, x + \Delta x]) = \int_x^{x+\Delta x} 1 dy = (x + \Delta x) - x = \Delta x$$

which justifies the name “uniform”



$U(a,b)$ density function.



10.2 RANDOM NUMBERS

◎ Exponential Distribution Function

Exponential	$\text{expo}(\beta)$
Possible applications	Times between independent events, e.g., arrivals at a service facility, that occur at a constant rate; lifetimes of devices with constant hazard rate
Density (see Fig. 5.2)	$f(x) = \begin{cases} \left(\frac{1}{\beta}\right) e^{-x/\beta} & \text{if } x \geq 0 \\ 0 & \text{otherwise} \end{cases}$
Distribution	$F(x) = \begin{cases} 1 - e^{-x/\beta} & \text{if } x \geq 0 \\ 0 & \text{otherwise} \end{cases}$
Parameter	Scale parameter $\beta > 0$
Range	$[0, \infty)$
Mean	β
Variance	β^2
Mode	0
MLE	$\hat{\beta} = \bar{X}(n)$

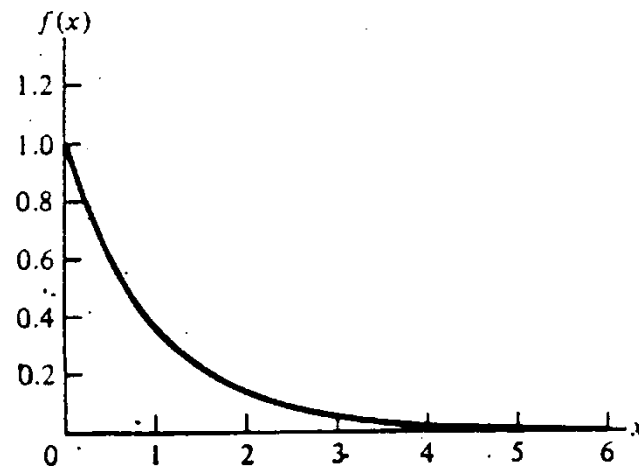


10.2 RANDOM NUMBERS

⊙ Exponential Distribution Function

Comments

1. The $\text{expo}(\beta)$ distribution is a special case of both the gamma and Weibull distributions (for shape parameter $\alpha = 1$ and scale parameter β in both cases)
2. If $X_1, X_2, \dots, X_m$ are independent $\text{expo}(\beta)$ random variables, then $X_1 + X_2 + \dots + X_m \sim \text{gamma}(m, \beta)$, also called the m -Erlang distribution
3. The exponential distribution is the only continuous distribution with the memoryless property (see Appendix 8A)



$\text{expo}(1)$ density function.

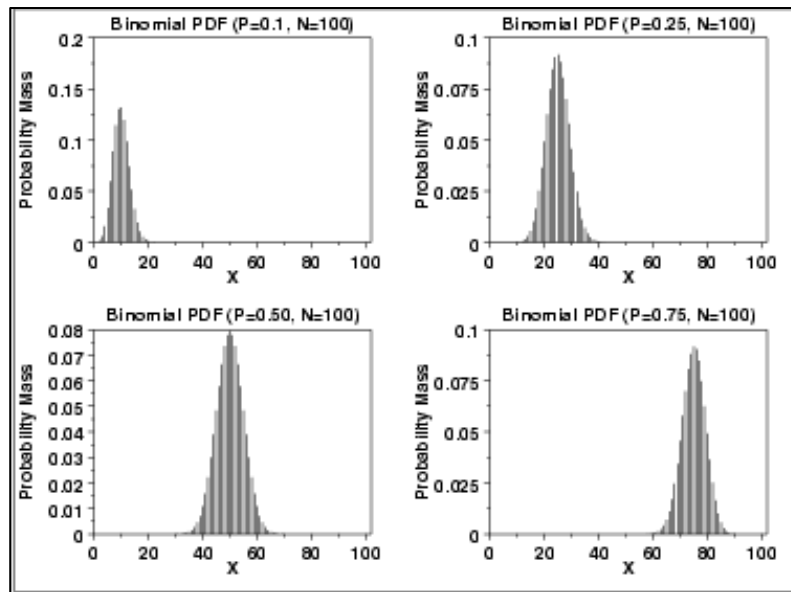


10.2 RANDOM NUMBERS

◎ Binomial Frequency Function

$$p(x) = C_x^m p^x q^{m-x}$$

$$E(x) = mp, \quad \text{Var}(x) = mpq$$

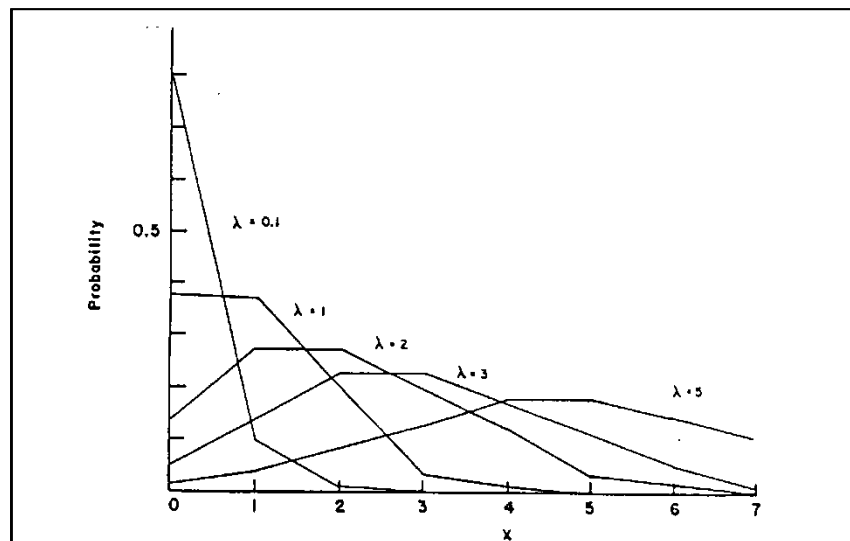


10.2 RANDOM NUMBERS

◎ Poisson Frequency Function

$$p(x) = \frac{\lambda^x}{x! e^\lambda} \quad \lambda = mp < 6$$

$$E(x) = \sum_{x=0}^{\infty} p(x)x = \lambda \quad \text{Var}(x) = \lambda$$

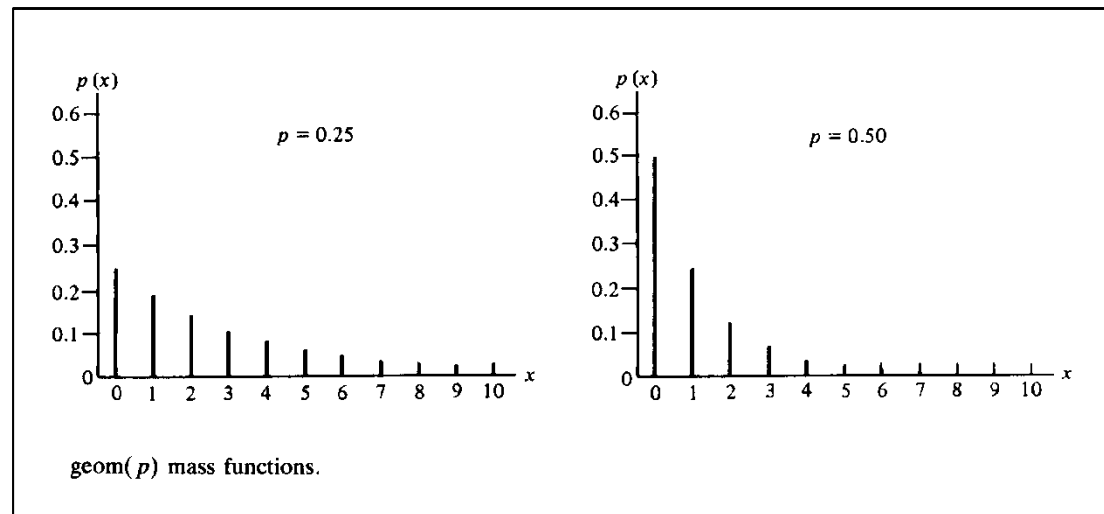


10.2 RANDOM NUMBERS

Geometric Distribution Function

$$p(x) = \begin{cases} p(1-p)^x & \text{if } x \in \{0, 1, \dots\} \\ 0 & \text{otherwise} \end{cases}$$

$$E(x) = \frac{1-p}{p} \quad \text{Var}(x) = \frac{1-p}{p^2}$$



10.2 RANDOM NUMBERS

Geometric	$\text{geom}(p)$
Possible applications	Number of failures before the first success in a sequence of independent Bernoulli trials with probability p of success on each trial; number of items inspected before encountering the first defective item; number of items in a batch of random size; number of items demanded from an inventory
Mass (see Fig. 6.14)	$p(x) = \begin{cases} p(1-p)^x & \text{if } x \in \{0, 1, \dots\} \\ 0 & \text{otherwise} \end{cases}$
Distribution	$F(x) = \begin{cases} 1 - (1-p)^{\lfloor x \rfloor + 1} & \text{if } x \geq 0 \\ 0 & \text{otherwise} \end{cases}$
Parameter	$p \in (0,1)$
Range	$\{0, 1, \dots\}$
Mean	$\frac{1-p}{p}$
Variance	$\frac{1-p}{p^2}$
Mode	0
MLE	$\hat{p} = \frac{1}{\bar{X}(n) + 1}$
Comments	1. If $Y_1, Y_2, \dots$ is a sequence of independent Bernoulli(p) random variables and $X = \min\{i: Y_i = 1\} - 1$, then $X \sim \text{geom}(p)$. 2. If $X_1, X_2, \dots, X_s$ are independent $\text{geom}(p)$ random variables, then $X_1 + X_2 + \dots + X_s$ has a negative binomial distribution with parameters s and p. 3. The geometric distribution is the discrete analog of the exponential distribution, in the sense that it is the only discrete distribution with the memoryless property (see Prob. 4.27) 4. The $\text{geom}(p)$ distribution is a special case of the negative binomial distribution (with $s = 1$ and the same value for p)



10.2 RANDOM NUMBERS

◎ Gamma Density Function

Gamma	$\text{gamma}(\alpha, \beta)$
Density (see Fig. 6.3)	$f(x) = \begin{cases} \frac{\beta^{-\alpha} x^{\alpha-1} e^{-x/\beta}}{\Gamma(\alpha)} & \text{if } x > 0 \\ 0 & \text{otherwise} \end{cases}$
Distribution	where $\Gamma(\alpha)$ is the <i>gamma function</i>, defined by $\Gamma(z) = \int_0^\infty t^{z-1} e^{-t} dt$ for any real number $z > 0$. Some properties of the gamma function: $\Gamma(z+1) = z\Gamma(z)$ for any $z > 0$, $\Gamma(k+1) = k!$ for any nonnegative integer k, $\Gamma(k + \frac{1}{2}) = \sqrt{\pi} \cdot 1 \cdot 3 \cdot 5 \cdots (2k-1)/2^k$ for any positive integer k, $\Gamma(1/2) = \sqrt{\pi}$ If α is not an integer, there is no closed form. If α is a positive integer, then $F(x) = \begin{cases} 1 - e^{-x/\beta} \sum_{j=0}^{\alpha-1} \frac{(x/\beta)^j}{j!} & \text{if } x > 0 \\ 0 & \text{otherwise} \end{cases}$
Parameters	Shape parameter $\alpha > 0$, scale parameter $\beta > 0$
Range	$[0, \infty)$
Mean	$\alpha\beta$
Variance	$\alpha\beta^2$
Mode	$\beta(\alpha - 1)$ if $\alpha \geq 1$, 0 if $\alpha < 1$



10.2 RANDOM NUMBERS

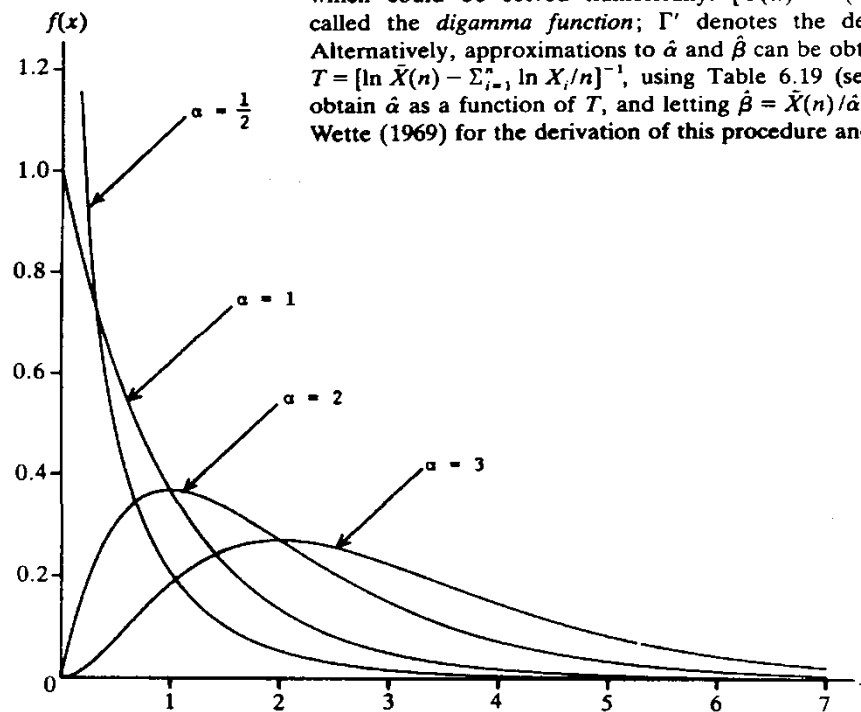
◎ Gamma Density Function

MLE

The following two equations must be satisfied:

$$\ln \hat{\beta} + \Psi(\hat{\alpha}) = \frac{\sum_{i=1}^n \ln X_i}{n}, \quad \hat{\alpha}\hat{\beta} = \bar{X}(n)$$

which could be solved numerically. [$\Psi(\hat{\alpha}) = \Gamma'(\hat{\alpha})/\Gamma(\hat{\alpha})$ and is called the *digamma function*; Γ' denotes the derivative of Γ .] Alternatively, approximations to $\hat{\alpha}$ and $\hat{\beta}$ can be obtained by letting $T = [\ln \bar{X}(n) - \sum_{i=1}^n \ln X_i/n]^{-1}$, using Table 6.19 (see App. 6A) to obtain $\hat{\alpha}$ as a function of T , and letting $\hat{\beta} = \bar{X}(n)/\hat{\alpha}$. [See Choi and Wette (1969) for the derivation of this procedure and of Table 6.19]



gamma(α ,1) density functions.



10.2 RANDOM NUMBERS

◎ Weibull Density Function

Weibull	Weibull(α, β)
Possible applications	Widely used in reliability models for lifetimes of devices; time to complete some task (density takes on shapes similar to gamma densities)
Density (see Fig. 5.4)	$f(x) = \begin{cases} \alpha\beta^{-\alpha}x^{\alpha-1}e^{-(x/\beta)^\alpha} & \text{if } x > 0 \\ 0 & \text{otherwise} \end{cases}$
Distribution	$F(x) = \begin{cases} 1 - e^{-(x/\beta)^\alpha} & \text{if } x > 0 \\ 0 & \text{otherwise} \end{cases}$
Parameters	Shape parameter $\alpha > 0$, scale parameter $\beta > 0$.
Range	$[0, \infty)$
Mean	$\frac{\beta}{\alpha} \Gamma\left(\frac{1}{\alpha}\right)$
Variance	$\frac{\beta^2}{\alpha} \left\{ 2\Gamma\left(\frac{2}{\alpha}\right) - \frac{1}{\alpha} \left[\Gamma\left(\frac{1}{\alpha}\right) \right]^2 \right\}$
Mode	$\begin{cases} \beta \left(\frac{\alpha-1}{\alpha} \right)^{1/\alpha} & \text{if } \alpha \geq 1 \\ 0 & \text{if } \alpha < 1 \end{cases}$



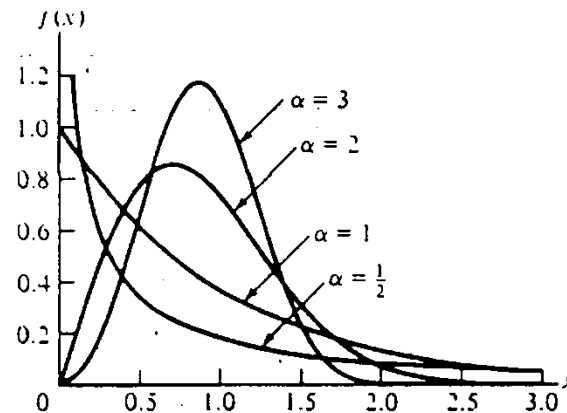
10.2 RANDOM NUMBERS

🎯 Weibull Density Function

Comments

1. The $\text{expo}(\beta)$ and $\text{Weibull}(1, \beta)$ distributions are the same
2. $X \sim \text{Weibull}(\alpha, \beta)$ if and only if $X^\alpha \sim \text{expo}(\beta^\alpha)$
3. The (natural) logarithm of a Weibull random variable has a distribution known as the *extreme-value* or *Gumbel distribution* [see [17], Mann [26], and part (b) of Prob. 7.1]
4. As $\alpha \rightarrow \infty$, the Weibull distribution becomes degenerate at β . Thus, Weibull densities for large α have a sharp peak at the mode
- 5.

$$\lim_{x \rightarrow 0} f(x) = \begin{cases} \infty & \text{if } \alpha < 1 \\ \frac{1}{\beta} & \text{if } \alpha = 1 \\ 0 & \text{if } \alpha > 1 \end{cases}$$



Weibull(α , 1) density functions.



10.2 RANDOM NUMBERS

◎ Beta Density Function

Beta	$\text{beta}(\alpha_1, \alpha_2)$
Possible applications	Used as a rough model in the absence of data (see Sec. 6.9); distribution of a random proportion, such as the proportion of defective items in a shipment; time to complete a task, e.g., in a PERT network
Density (see Fig. 6.7)	$f(x) = \begin{cases} \frac{x^{\alpha_1-1}(1-x)^{\alpha_2-1}}{B(\alpha_1, \alpha_2)} & \text{if } 0 < x < 1 \\ 0 & \text{otherwise} \end{cases}$ where $B(\alpha_1, \alpha_2)$ is the <i>beta function</i>, defined by $B(z_1, z_2) = \int_0^1 t^{z_1-1}(1-t)^{z_2-1} dt$ for any real numbers $z_1 > 0$ and $z_2 > 0$. Some properties of the beta function: $B(z_1, z_2) = B(z_2, z_1), \quad B(z_1, z_2) = \frac{\Gamma(z_1)\Gamma(z_2)}{\Gamma(z_1 + z_2)}$
Distribution	No closed form, in general. If either α_1 or α_2 is a positive integer, a binomial expansion can be used to obtain $F(x)$, which will be a polynomial in x , and the powers of x will be, in general, positive real numbers ranging from 0 through $\alpha_1 + \alpha_2 - 1$
Parameters	Shape parameters $\alpha_1 > 0$ and $\alpha_2 > 0$
Range	$[0, 1]$
Mean	$\frac{\alpha_1}{\alpha_1 + \alpha_2}$
Variance	$\frac{\alpha_1 \alpha_2}{(\alpha_1 + \alpha_2)^2 (\alpha_1 + \alpha_2 + 1)}$
Mode	$\begin{cases} \frac{\alpha_1 - 1}{\alpha_1 + \alpha_2 - 2} & \text{if } \alpha_1 > 1, \alpha_2 > 1 \\ 0 \text{ and } 1 & \text{if } \alpha_1 < 1, \alpha_2 < 1 \\ 0 & \text{if } (\alpha_1 < 1, \alpha_2 \geq 1) \text{ or if } (\alpha_1 = 1, \alpha_2 > 1) \\ 1 & \text{if } (\alpha_1 \geq 1, \alpha_2 < 1) \text{ or if } (\alpha_1 > 1, \alpha_2 = 1) \\ \text{does not uniquely exist} & \text{if } \alpha_1 = \alpha_2 = 1 \end{cases}$



10.2 RANDOM NUMBERS

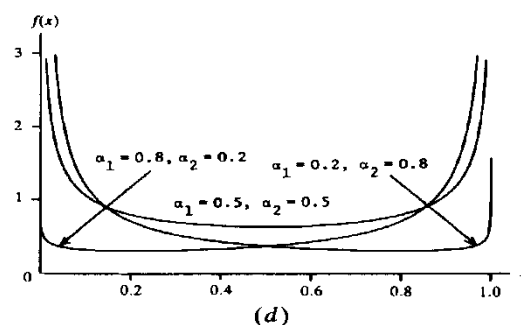
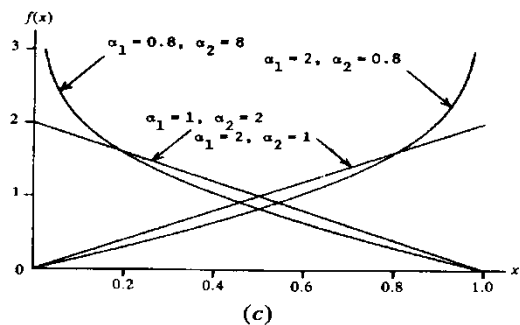
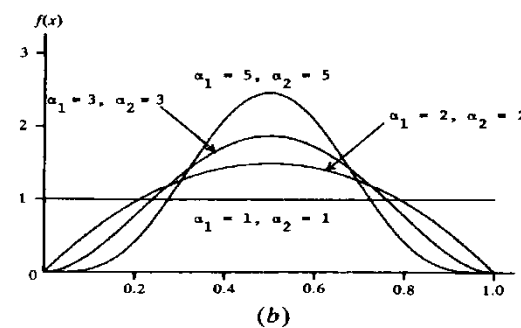
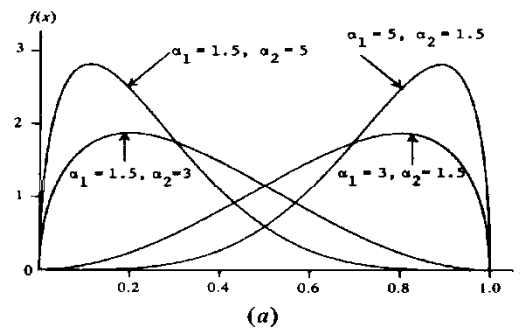
◎ Beta Density Function

MLE

The following two equations must be satisfied:

$$\Psi(\hat{\alpha}_1) - \Psi(\hat{\alpha}_1 + \hat{\alpha}_2) = \ln G_1, \quad \Psi(\hat{\alpha}_2) - \Psi(\hat{\alpha}_1 + \hat{\alpha}_2) = \ln G_2$$

where Ψ is the digamma function, $G_1 = (\prod_{i=1}^n X_i)^{1/n}$, and $G_2 = [\prod_{i=1}^n (1 - X_i)]^{1/n}$ [see Gnanadesikan, Pinkham, and Hughes (1967)]; note that $G_1 + G_2 \leq 1$. These equations could be solved numerically [see Beckman and Tietjen (1978)], or approximations to $\hat{\alpha}_1$ and $\hat{\alpha}_2$ can be obtained from Table 6.20 (see App. 6A), which was computed for particular (G_1, G_2) pairs by modifications of the methods in Beckman and Tietjen (1978)



10.3 SAMPLING STRATEGIES

- ◎ **Random Sampling:** This is the simplest strategy. With sufficient number of draws, this method will produce a sample of selected values whose distribution resembles that of the original. This method is inefficient in generating deviates from the tails of the distribution.
- ◎ **Stratified Sampling:** A strategy to protect against potential small-sample bias by classifying the population by the relevant groups or categories and then select randomly from each of these sub-populations.
- ◎ **Latin Hypercube Sampling (LHS):** LHS is similar to stratified sampling, but is based more explicitly on the underlying probability distributions.



10.3 SAMPLING STRATEGIES

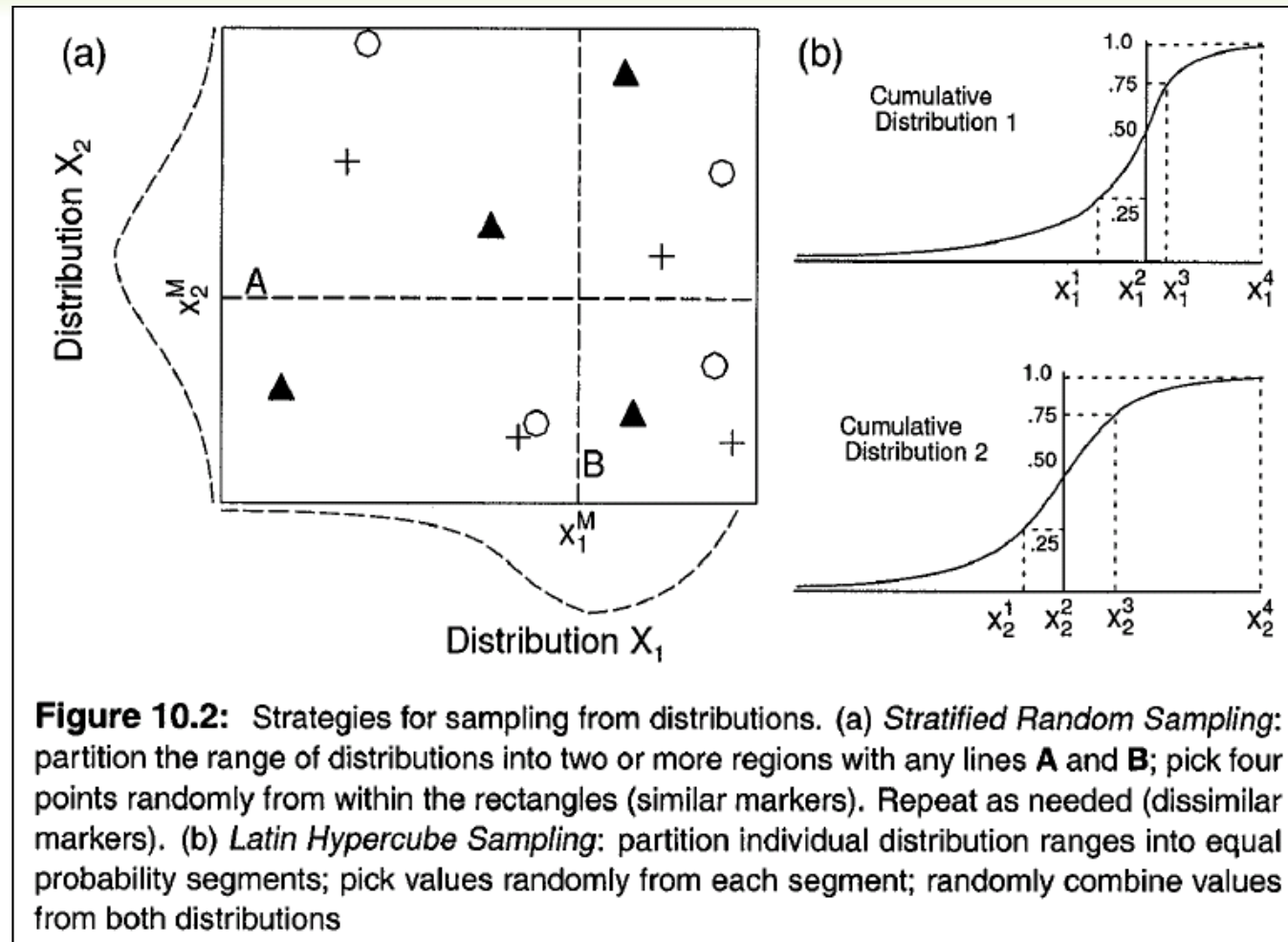


Figure 10.2: Strategies for sampling from distributions. (a) *Stratified Random Sampling*: partition the range of distributions into two or more regions with any lines **A** and **B**; pick four points randomly from within the rectangles (similar markers). Repeat as needed (dissimilar markers). (b) *Latin Hypercube Sampling*: partition individual distribution ranges into equal probability segments; pick values randomly from each segment; randomly combine values from both distributions



10.4 APPLICATIONS TO DIFFERENTIAL EQUATIONS

- ◎ Randomness may be implemented in differential equation models in the initial conditions, driving variables, parameters, or on the state variables directly.
- ◎ The most common application is to randomize driving variables and parameters.
- ◎ There are two different concepts of biological stochasticity: *environmental* and *demographic stochasticity*. Environmental stochasticity refers to random variation in systems modeled as populations or compartments. Demographic stochasticity refers to random variation in the occurrence of events affecting the state of an individual.



10.4 APPLICATIONS TO DIFFERENTIAL EQUATIONS

- ◎ To incorporate stochastic events in parameter values, we use a differential equation in which the parameters at time t are affected by random deviates from some distribution (e.g., normal). For example, a stochastic density-independent population model might be

$$\frac{dX}{dt} = r_t[N(\mu, \sigma^2)]X \quad (10.5)$$



10.4 APPLICATIONS TO DIFFERENTIAL EQUATIONS

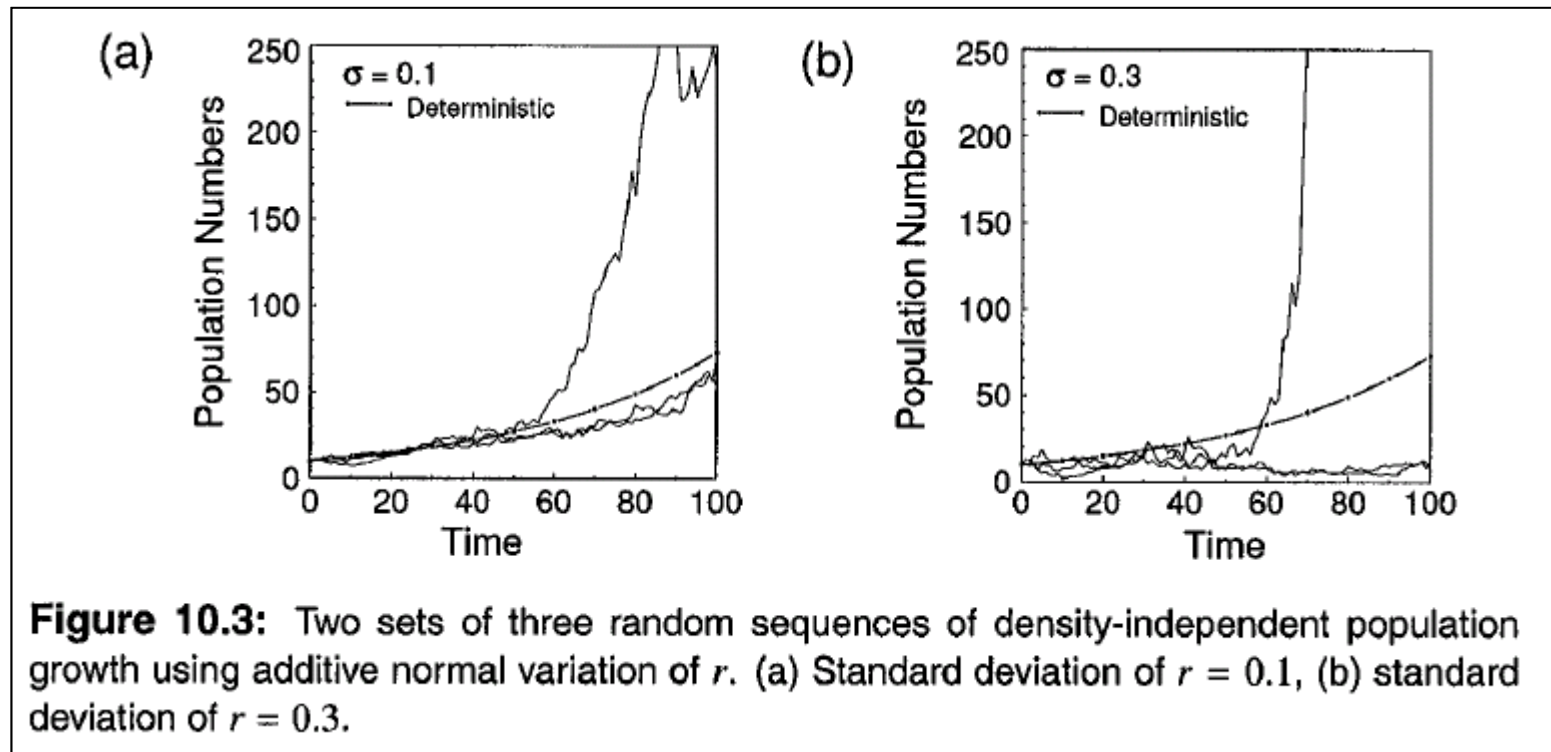
◎ Steps for the Simulation:

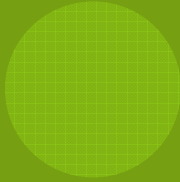
1. Determine the probability distribution to use for the parameter and estimate the descriptive statistics (mean and variance).
2. Inside the simulation loop, sample the distribution and use the resulting random deviate as the parameter value in the differential equation.
3. Save the resulting dynamics in an array for post-simulation statistical analysis.
4. Repeat steps 2 and 3 a large number of times to obtain a set of Monte Carlo replicates on which to do statistics. The size of "large" depends on the question being addressed, the underlying variability of the biological process, and the amount of time and money available to answer the question, but 10,000 replicates is not uncommon.
5. Perform statistical analysis on the resulting random dynamics.



10.4 APPLICATIONS TO DIFFERENTIAL EQUATIONS

- Example of a stochastic density-independent population model





10.4 APPLICATIONS TO DIFFERENTIAL EQUATIONS

- ◎ Some Questions in Analyzing Random System Dynamics
 - ◎ What is the nature of the state variable values of a single system subject to environmental stochasticity?
 - ◎ What is the nature of the statistical distribution of an *ensemble of systems*, where each is subject to similar environmental stochasticity?
 - ◎ What are the stability properties of the random dynamics?



10.4 APPLICATIONS TO DIFFERENTIAL EQUATIONS

- ◎ An Example of Stochastic Logistic Equation (May 1973):

$$\frac{dV}{dt} = rV \left(1 - \frac{V}{K} \right)$$

Eliminating r by dividing both sides by r/K

$$\frac{dV}{dt(r/K)} = V(K - V)$$

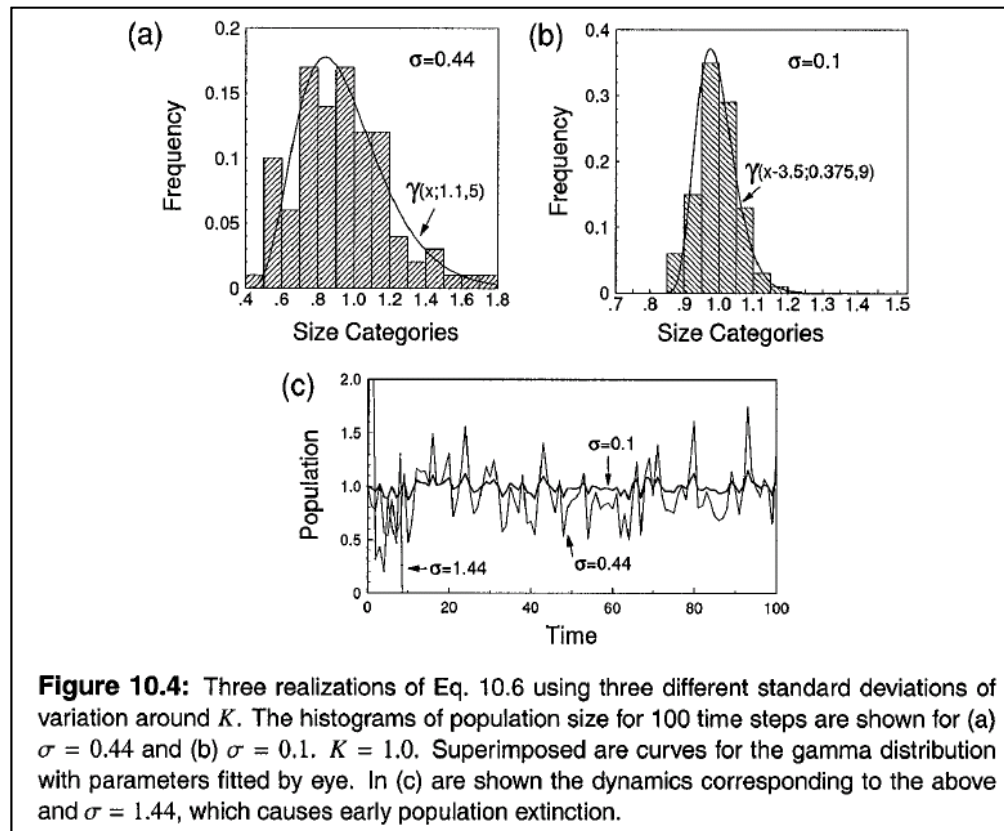
Defining a new time variable $\tau = t(r/K)$ and an additive model of stochastic variation in K

$$\frac{dV}{d\tau} = V(K + N(0, \sigma^2) - V) \quad (10.6)$$



10.4 APPLICATIONS TO DIFFERENTIAL EQUATIONS

◎ An Example of Stochastic Logistic Equation (May 1973):



10.5 MARKOV PROCESSES

- ◎ A Markov process is a probabilistic model of system dynamics when the system variables possess only a finite number of possible states.
- ◎ The rules that describe the changes can be either deterministic or probabilistic; normally, we are interested in the latter form.
- ◎ There are two possible ways of viewing the system. First, we may think of the system as an individual object, in which case the system visits the various states sequentially. Second, we may envision the system to be an ensemble of individuals that are not explicitly modeled, but each of which is viewed as changing state randomly.



10.5 MARKOV PROCESSES

- Both approaches can be described with the same mathematics. The central concept is the transition matrix. A transition matrix is a special case of a probability matrix which is an $n \times n$ matrix in which all elements are non-negative, and the elements in the rows sum to 1.0. For example,

$$\mathbf{P} = \begin{pmatrix} \frac{1}{4} & \frac{1}{4} & \frac{1}{2} \\ 0 & 1 & 0 \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \end{pmatrix} \quad (10.7)$$

- Two important facts of these matrices are: (1) If $\mathbf{P}$ and $\mathbf{Q}$ are probability matrices, then $\mathbf{PQ}$ is a probability matrix. (2) If $\mathbf{P}$ is a probability matrix, then there is a row vector $\mathbf{t}$ such that

$$\mathbf{t} = \mathbf{tP} \quad (10.8)$$



10.5 MARKOV PROCESSES

- ◎ Biological Applications of Markov Processes
 - ◎ Since $\mathbf{P}$ is the probability of moving from the current state to the next state, it is convenient to call this the one-step transition probability matrix (Hillier and Lieberman 1980).
 - ◎ $\mathbf{P}$ multiplied by itself ($\mathbf{P}^{(2)}$) is the two-step transition probability matrix and represents the probabilities of moving from state i to state j in two steps. $\mathbf{P}^{(n)}$ is defined similarly for n steps.
 - ◎ Let $\mathbf{p}$ be a row vector of probabilities that an individual is in state i . In the ensemble interpretation, it is the fraction of individuals in state i . Then, we can form a recursive equation to generate the probability distribution in the next time step as

$$\mathbf{p}_{t+1} = \mathbf{p}_t \mathbf{P} \quad (10.9)$$



10.5 MARKOV PROCESSES

- ◎ Biological Applications of Markov Processes
 - ◎ If $\mathbf{P}$ is composed entirely of elements that are constant and independent of p_i and if $\mathbf{p}_{t+1}$ depends only on $\mathbf{p}_t$, (i.e., not on previous $\mathbf{p}_{t-m}$ where m is a positive integer), then $\mathbf{P}$ is a *Markov transition matrix*. In this case, Eq. 10.9 describes a Markov process. Sometimes this is referred to as a linear; first-order Markov process.
 - ◎ As stated above, for a given $\mathbf{P}$, there is a $\mathbf{p}$ such that in Eq. 10.9, $\mathbf{p}_{t+1} = \mathbf{p}_t$. This is, basically, an equilibrium of the probability distribution of the system, and $\mathbf{p}_t$ is called the *fixed probability distribution*. The fixed $\mathbf{p}$ can be computed from

$$\mathbf{p} = \mathbf{p}_0 \mathbf{P}^{(n)}$$



10.5 MARKOV PROCESSES

© Biological Applications of Markov Processes

Table 10.1: The Markov transition probability matrix for a deer moving among water, grass, and sleeping areas. The powers of the matrices are indicated by the superscripts: 1, 2, 4, 16, 32, 64.

$\mathbf{p}^1$			$\mathbf{p}^2$		
0.600000	0.200000	0.200000	0.460000	0.270000	0.270000
0.250000	0.500000	0.250000	0.337500	0.362500	0.300000
0.250000	0.250000	0.500000	0.337500	0.300000	0.362500
$\mathbf{p}^4$			$\mathbf{p}^{16}$		
0.393850	0.303075	0.303075	0.384754	0.307623	0.307623
0.378844	0.312531	0.308625	0.384529	0.307743	0.307728
0.378844	0.308625	0.312531	0.384529	0.307728	0.307743
$\mathbf{p}^{32}$			$\mathbf{p}^{64}$		
0.384616	0.307692	0.307692	0.384616	0.307692	0.307692
0.384615	0.307692	0.307692	0.384616	0.307692	0.307692
0.384615	0.307692	0.307692	0.384616	0.307692	0.307692



10.5 MARKOV PROCESSES

- ◎ Simulating Markov and Transition Matrix Models
 - ◎ The assumptions of a Markov process are biologically unrealistic. In particular, the current state of the system will often influence the transition probabilities, so that $\mathbf{P}$ will not be constant.
 - ◎ One approach to relaxing these assumptions is to use semi-Markov processes applied to compartments (e.g., spatial position) in which the probability of leaving increases the longer an object has been in the compartment.



10.5 MARKOV PROCESSES

◎ Simulating Markov and Transition Matrix Models

- ◎ The simulation process can be simplified if the rows of the original transition matrix are converted to cumulative distributions. Then we can use table look-up on an empirical distribution. The rows denote the current state; the columns denote the new states. Given the transformed transition matrix ($\mathbf{P}'$), the algorithm is:

1. Assign an initial state to the system ($s_{i,t}$).
2. Obtain a uniform random deviate (U_t).
3. For row $s_{i,t}$, determine the column ($s_{j,t+1}$) such that $p_{ij} < U_t \leq p_{i(j+1)}$ where p_{ij} is the upper bound of the cumulative distribution for the transition from state i to state j .
4. j is the new $s_{j,t+1}$.

